

COMMAND NEXUS · CIT 140 · WEEK 1

Problem Brief

Meridian Forge Robotics · The Process Below

LEARNER Christian Calkins	ARTIFACT Problem Brief	STATUS Working document
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SECTION 01

Visible Failure

Test began at 2:13:07
Repeat cycle begins
Stop request returned: ACTIVE_INSTRUCTION_PENDING
Nia reports that MR-17 behaved normally before the incident.
Elias says a controlled restart may clear the active state.
Mara confirms production has paused in the affected QA cell

SECTION 02

Verified Facts

The robot is not scheduled CURRENTLY. (Nia)

Within the document it states that MR-17 is not assigned to do anything.

Mara states that production is paused on the QA cell.

SECTION 03

Assumptions and Open Questions

Nia reports that MR-17 was functionally operable, what happened between the time of the error and proper function? (Question)

2. I would need to see the logs and come to the conclusion that MR-17 was working beforehand. (Assumption)

SECTION 04

System Environment Map

SECTION 04A

People and Roles

Elias cannot continue his work, he was there when the system was negatively impacted.

Nia reported that the system was fine before the incident occurred.

Mara. Mara knows that everyone is affected.

SECTION 04B

Processes and Rules

Mara reported that the production on the facility in the QA cell is affected by the productivity MR-17. Depending on the functionality of the MR-17 robot, Elias can look into the robot's code. I can provide my input and make the robot restart, or provide another solution.

SECTION 04C

Information and Data

Mara states that production is paused on the QA cell.

Nia reports that MR-17 was functionally operable

The source label says legacy_sync.exe.

The command was validly formatted,

Nia reports that MR-17 behaved normally before the incident.

Elias says a controlled restart may clear the active state.

The robot is not scheduled CURRENTLY. (Nia)

SECTION 04D

Technology and Tools

MR-17 Format accepts the command according to Elias. It runs through legacy_sync.exe

Elias is unfamiliar with legacy_sync.exe.

The telemetry doesn't tell us who owns legacy_sync or prove why MR-17 is not working.

The telemetry and .exe work but during the incident MR-17 stopped working after the command was input.

SECTION 04E

Environment and Constraints

The robot not working, the inability to understand why, and the ability to fix MR-17.

SECTION 05

Investigation Boundary

Inside the Investigation

The team will examine the robot for any traces of the command that caused the robot to malfunction. They need to study the error and determine where it came from.

Outside for Now

Current evidence doesn't require the robot to work. We know that it worked previously before legacy_sync.exe caused the incident. We know that the command fit the format that is accepted. And so, we don't need this.

SECTION 06

Organizational Impact

MR-17 is halted. Production is stopped, therefore MR-17 affects the entire facility negatively. Mara provides evidence that the QA cell is halted. This prevents work being done in the lab.

SECTION 07

Problem Statement

The robot is broken after a command labeled legacy_sync.exe is ran. This causes MR-17 to halt entirely. While it wasn't scheduled, it shuts down the entire sector, preventing any progress in the QA cell. The gap is the inability to figure out what is causing the problem and how to possibly remedy it to get the progress going and MR-17 to work properly like it was before the command.

SECTION 08

Next Investigation Step

“Should we restart the robot now?”

Investigation:

Document the existence of the command. Run different commands upon restart to see if you can duplicate the effect. Restarting the robot would ensure that we find out if it will be fixed. The restart will wipe the memory of the command that executed before the robot began to malfunction. The solution would be: Keep an eye on the robot after the restart and watch for legacy_sync.exe. source label.

SHARED ARTIFACT

Working Document Link

<https://github.com/chris031903/CIT140-Portfolio.git>

WEEK 1 STOPPING RULE

Define before designing.

This Problem Brief records the problem that requires investigation. It does not recommend a solution.